

S5.6 Mathematics Mock Exam - Semester 2 2026

Student Name: _____

Date: _____

Class: _____

Instructions:

- Answer all questions in your separate notebook.
 - Show all necessary working clearly.
 - Give your answers to 3 significant figures unless specified otherwise.
 - Calculator use is permitted.
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Topic 1: Quadratics and the Parabola

Question 1

Consider the quadratic function $y = 2x^2 - 12x + 10$.

a) Write this quadratic function in the vertex form $y = a(x - p)^2 + q$.

(3 Marks — Knowledge: 2, Application: 1)

b) Using your result from part (a), state: i) The coordinates of the vertex of the parabola. ii) The equation of the line of symmetry.

(2 Marks — Knowledge: 1, Communication: 1)

c) Find the roots (zeros) of the parabola by solving $y = 0$.

(2 Marks — Knowledge: 1, Application: 1)

Question 2

Solve the quadratic inequality $x^2 - 5x - 6 < 0$. Find the region of solutions for x and show your solution using interval notation.

(4 Marks — Knowledge: 2, Application: 2)

Topic 2: Log Equations

Question 3

Solve the following logarithmic equation for x :

$$\log_2(x) + \log_2(x - 2) = 3$$

Show all your working and check for any extraneous solutions.

(4 Marks — Knowledge: 2, Application: 2)

Question 4

Using the laws of logarithms, solve the following equation for x :

$$2 \log_3(x) - \log_3(x + 6) = 1$$

(4 Marks — Knowledge: 2, Application: 2)

Topic 3: Trigonometry

Question 5

A surveyor is measuring a triangular piece of land PAB . The angle at vertex P is 65° . The distance from P to A is 8 m, and the distance from P to B is 12 m.

a) Calculate the distance from A to B , giving your answer correct to 3 significant figures.

(3 Marks — Application: 2, Communication: 1)

b) Find the area of the triangular piece of land PAB , correct to 1 decimal place.

(2 Marks — Knowledge: 1, Application: 1)

Question 6

a) Solve the trigonometric equation $2 \sin(2x - \frac{\pi}{3}) = \sqrt{3}$ for $0 \leq x \leq \pi$.

(3 Marks — Knowledge: 1, Application: 2)

b) Solve the quadratic trigonometric equation $2 \cos^2(x) - \sin(x) - 1 = 0$ for $0 \leq x \leq 2\pi$.

Topic 4: Probability

Question 7

In a group of 50 students, 30 study Mathematics, 25 study Physics, and 10 study both.

a) Construct a Venn diagram to represent this information.

(3 Marks — Knowledge: 1, Communication: 2)

b) A student is chosen at random. Find the probability that they study Mathematics, given that they study Physics.

(2 Marks — Knowledge: 1, Application: 1)

c) Find the probability that they do not study Mathematics, given that they study Physics.

(2 Marks — Knowledge: 1, Application: 1)

Question 8

A box contains 5 red balls and 3 green balls. Two balls are drawn at random from the box, one after another, without replacement.

a) Draw a tree diagram to represent this probability process, showing all branch probabilities clearly.

(3 Marks — Knowledge: 1, Communication: 2)

b) Calculate the probability that the two balls drawn are of different colors.

(3 Marks — Knowledge: 1, Application: 2)

Topic 5: Vectors in 2D

Question 9

Let the vectors $\vec{u}$ and $\vec{v}$ be defined as:

$$\vec{u} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}, \quad \vec{v} = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$$

a) Calculate the resultant vector $\vec{w} = 3\vec{u} + 2\vec{v}$.

(2 Marks — Knowledge: 1, Application: 1)

b) Calculate the magnitude of vector $\vec{u}$ and the magnitude of vector $\vec{v}$.

(3 Marks — Knowledge: 2, Application: 1)

Question 10

Given two vectors $\vec{a} = 3\vec{i} + 4\vec{j}$ and $\vec{b} = 12\vec{i} - 5\vec{j}$:

a) Calculate the scalar product (dot product) $\vec{a} \cdot \vec{b}$.

(2 Marks — Knowledge: 2)

b) Find the angle θ between vectors $\vec{a}$ and $\vec{b}$. Give your answer correct to the nearest degree.

(3 Marks — Knowledge: 1, Application: 2)

Topic 6: 3D Shapes

Question 11

A cuboid has dimensions $AB = 3$ cm, $BC = 4$ cm, and height $CG = 12$ cm.

a) Calculate the length of the diagonal of the base, AC .

(2 Marks — Knowledge: 1, Application: 1)

b) Calculate the length of the space diagonal of the cuboid, AG .

(2 Marks — Knowledge: 1, Application: 1)

Question 12

a) A solid cylinder has a radius of 3 cm and a height of 8 cm. Calculate the total surface area of the cylinder, leaving your answer in terms of π .

(3 Marks — Knowledge: 1, Application: 2)

b) A right pyramid has a square base of side length 6 cm and a vertical height of 10 cm. Calculate the volume of the pyramid.

(2 Marks — Knowledge: 1, Application: 1)